

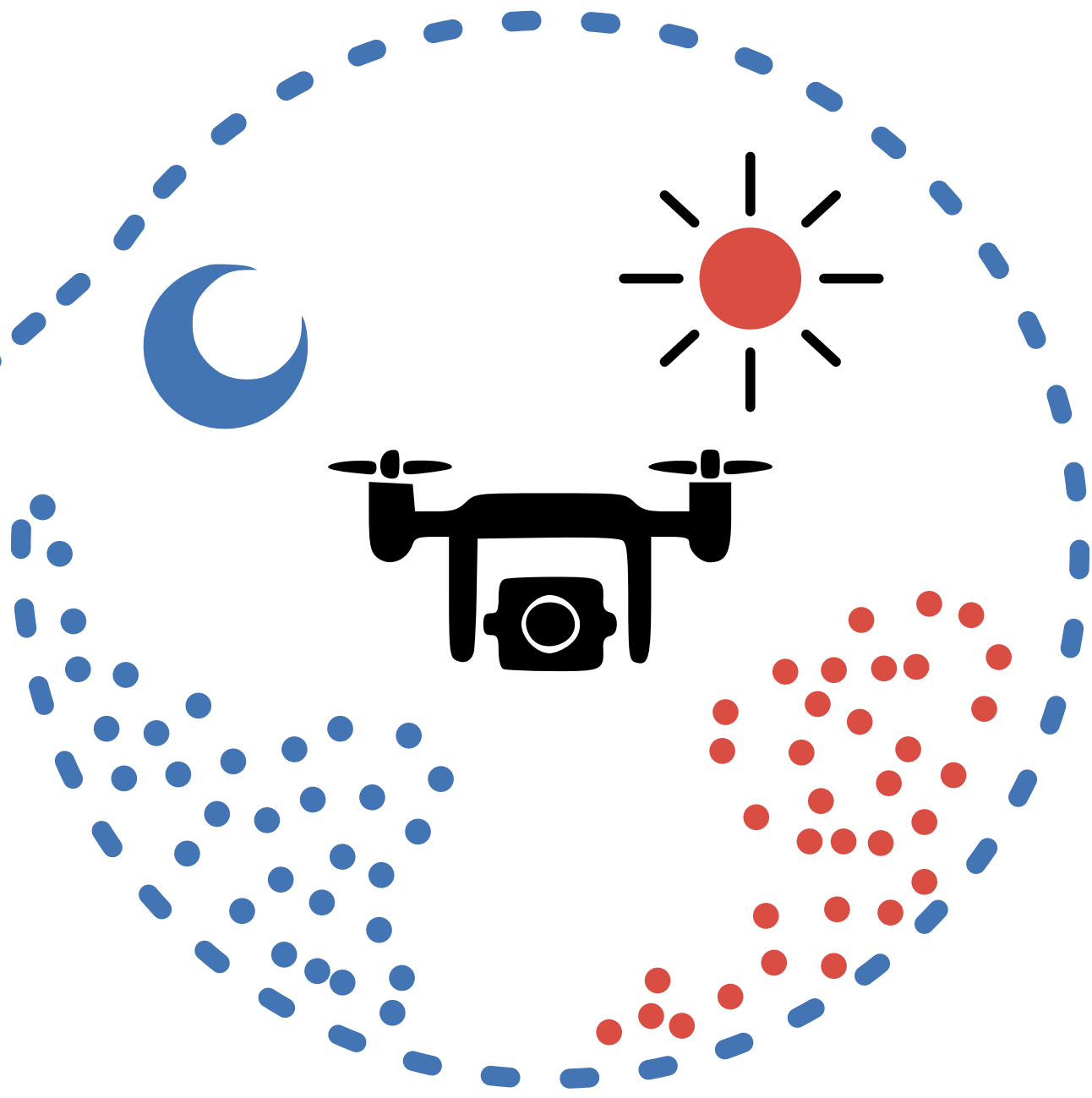
Should UAV-lidar be collected at night?

Impacts of Solar Illumination on Intensity, Penetration, and Ground Surface Detection Over Dense and Wet Vegetation

Introduction

UAV lidar is sensitive to environmental conditions that affect signal strength, vegetation penetration, and classification accuracy.

Although nighttime flights reduce solar noise, humidity, dew, and fog may offset these gains. This study examines how illumination influence UAV lidar data quality in peatland vegetation.

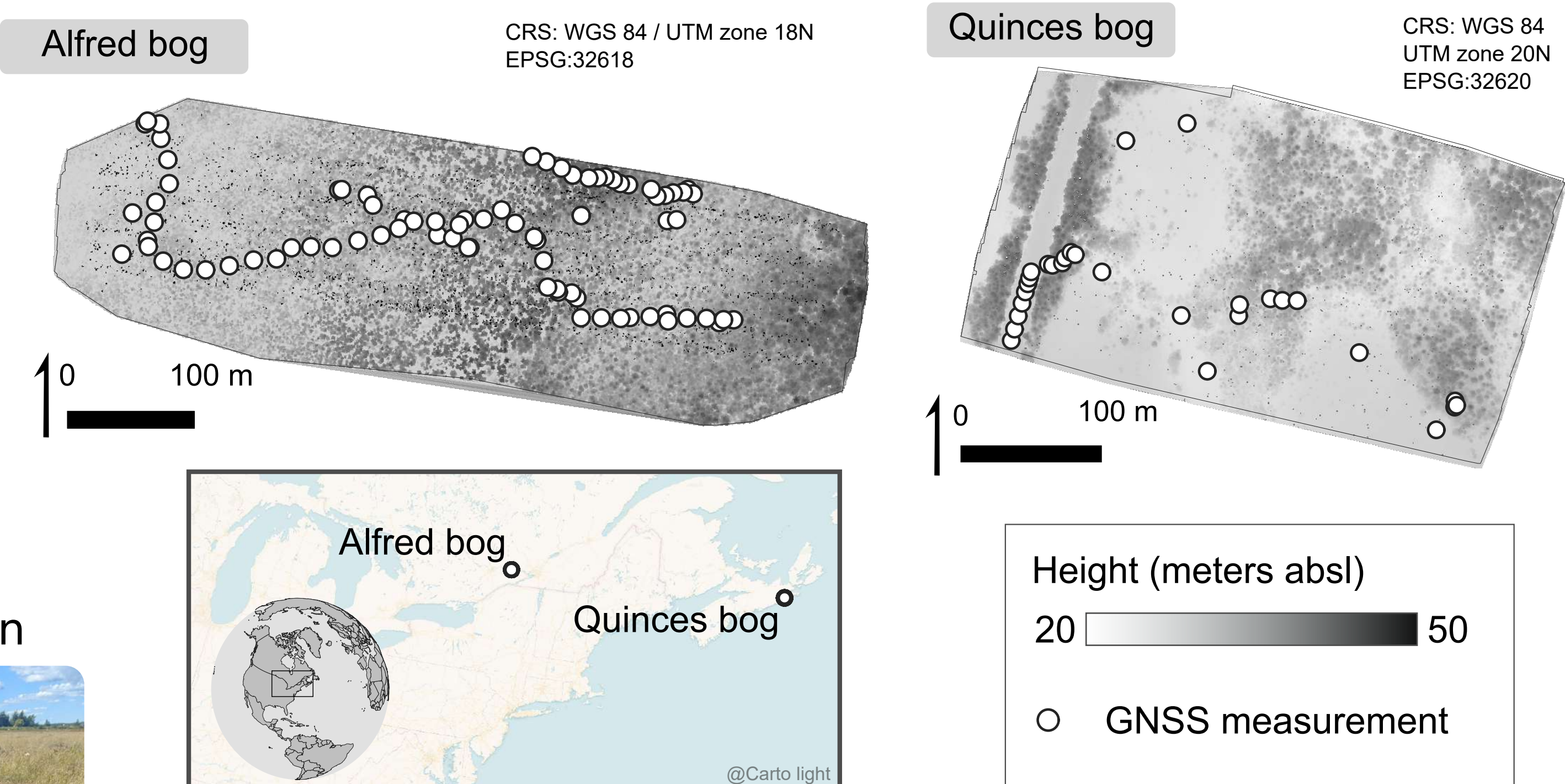


Hypothesis

- H1: Nighttime UAV lidar **improves data quality** compared with daytime acquisition under variable vegetation and surface wetness conditions.
- H2: Nighttime acquisition enables UAV lidar to maintain comparable point-cloud quality at higher flight altitudes than daytime acquisition.

Data

- 12 UAV missions across:
2 bog sites × 3 flight altitudes × 2 acquisition periods
Alfred/Quinces × 40/75/100 m × day/night
- 147 ground reference points
- Three surface canopy conditions



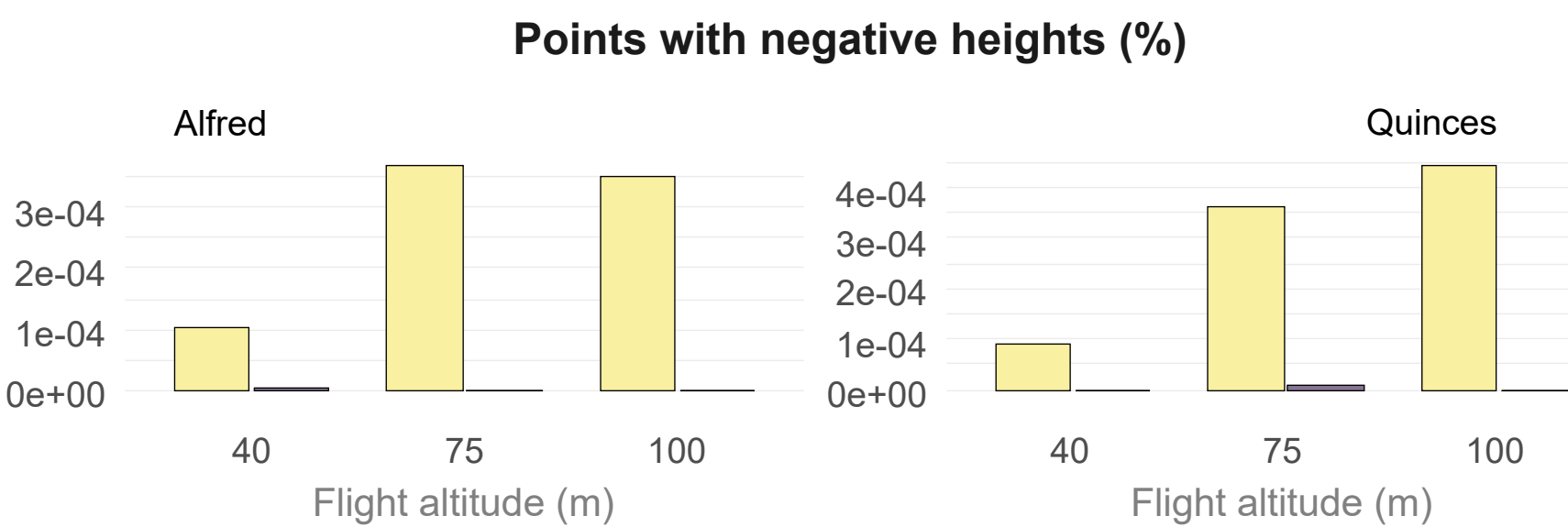
Night-flight advantages

Methods

1. Define ROI
Setting a common flight mission area in each bog site
2. Retile files
Create an efficient processing grid
3. Compute ROI metrics
4. Classify ground points
Progressive TIN Densification (PTM)
5. Compute target metrics
Use ground points within 50 cm radius and GNSS reference height data



Fewer negative height deviations at night.

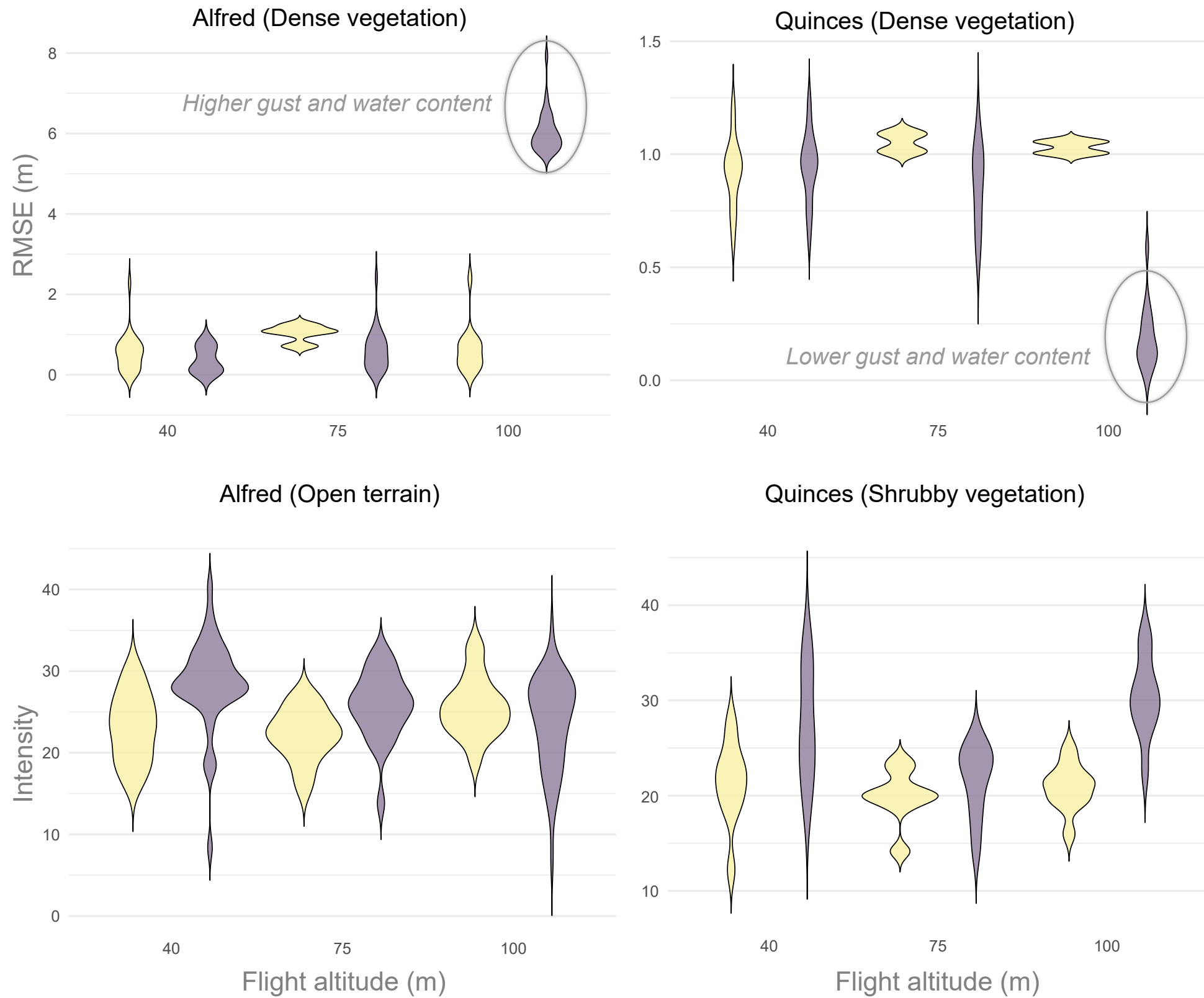


More second and third returns at night.



day night

Equal or lower RMSE error at night



Results and conclusion